

Comparison of Catalytic Activity and Inhibitors of Quinone Reactions of Succinate Dehydrogenase (Succinate–Ubiquinone Oxidoreductase) and Fumarate Reductase (Menaquinol–Fumarate Oxidoreductase) from *Escherichia coli*¹

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Escherichia coli succinate–ubiquinone oxidoreductase (SQR) and menaquinol–fumarate reductase (QFR) are excellent model systems to understand the function of eukaryotic Complex II. They have structural and catalytic properties similar to their eukaryotic counterpart. An exception is that potent inhibitors of mammalian Complex II, such as thenoyltrifluoroacetone and carboxanilides, only weakly inhibit their bacterial counterparts. This lack of good inhibitors of quinone reactions and the higher level of side reactions in the prokaryotic enzymes has hampered the elucidation of the mechanism of quinone oxidation/reduction in *E. coli* Complex II. In this communication DT-diaphorase and an appropriate quinone are used to measure quinol–fumarate reductase activity and *E. coli* *bo*-oxidase and quinones are used to determine succinate–quinone reductase activity. Simple Michaelis kinetics are observed for both enzymes with ubiquinones and menaquinones in the succinate oxidase (forward) and fumarate reductase (reverse) reactions. The comparison of *E. coli* SQR and QFR demonstrates that 2-*n*-heptyl 4-hydroxyquinoline-*N*-oxide (HQNO) is a potent inhibitor of QFR in both assays; however, SQR is not sensitive to HQNO. A series of 2-alkyl-4,6-dinitrophenols and pentachlorophenol were found to be potent competitive inhibitors of both SQR and QFR. In addition, the isolated *E. coli* SQR complex demon-

strates a mixed-type inhibition with carboxanilides, whereas the QFR complex is resistant to this inhibitor. The kinetic properties of SQR and QFR suggest that either ubiquinone or menaquinone operates at a single exchangeable site working in forward or reverse reactions. The pH activity profiles for *E. coli* QFR and SQR are similar showing maximal activity between pH 7.4 and 7.8, suggesting the importance of similar catalytic groups in quinol deprotonation and oxidation.

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Key Words: succinate–quinone oxidoreductase; succinate dehydrogenase; menaquinol–fumarate oxidoreductase; fumarate reductase; quinone inhibitors; Complex II.

Complex II refers collectively to succinate–ubiquinone oxidoreductase (SQR)³ or menaquinol–fumarate oxidoreductase (QFR), membrane-bound enzyme complexes that are similar in both structure and function. SQR couples the oxidation of succinate to fumarate in the Krebs cycle and transfers reducing equivalents to ubiquinone as part of the aerobic respiratory chain, whereas QFR oxidizes menaquinol and reduces fuma-

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³ Abbreviations used: SQR, succinate–ubiquinone oxidoreductase; QFR, menaquinol–fumarate oxidoreductase; HQNO, 2-*n*-heptyl-4-hydroxyquinoline-*N*-oxide; DCIP, 2,6-dichlorophenolindophenol; PES, phenazine ethosulfate; carboxin, 5,6-dihydro-2-methyl-1,4-oxathiin-3-carboxanilide; DNP, 2-alkyl-4,6-dinitrophenol substituted in 2-position; Q₁ and Q₂, homologs of ubiquinone having one or two isoprenoid units at position 6 of the quinone ring, respectively; MQ₁, menaquinone-1; DMN, 2,3-dimethyl-1,4-naphthoquinone; FAD, flavin adenine dinucleotide; PCP, pentachlorophenol.

rate to succinate as part of an anaerobic respiratory chain (1–3). Complex II from most organisms contains four subunits with a large highly conserved domain extrinsic to the membrane composed of a flavoprotein subunit (~66 kDa) containing a covalently bound FAD cofactor and the dicarboxylate binding site (2–4) and an iron–sulfur protein subunit (~27 kDa) containing three distinct iron–sulfur clusters (5, 6). This domain is bound to the membrane anchor domain through two small integral membrane subunits (~13–15 kDa) that are necessary to form the quinone binding site(s) found in both enzyme complexes (7–10). The membrane anchor domain shows the largest variation among species sometimes consisting of a single larger subunit (~23–30 kDa) and contains either one or two protoheme IX groups, with hexa-coordinated iron or no heme at all (3).

Escherichia coli succinate dehydrogenase and fumarate reductase are excellent model systems for studying the properties of Complex II. In *E. coli* two distinct operons encoding the SQR (*sdhCDAB*) and QFR (*frdABCD*) subunits are found (2, 3). It has been shown that the SQR or QFR gene products can functionally replace each other if conditions are provided that allow expression of the operons either aerobically or anaerobically (11, 12). Although QFR and SQR from *E. coli* are highly similar, one difference is that SQR contains an additional prosthetic group, a heme b_{556} cofactor, which bridges the two membrane anchor subunits (SdhC and SdhD) (13), whereas *E. coli* QFR is an example of a Complex II that lacks heme. There are many conflicting data in the literature regarding the turnover number for SQR and QFR from different eukaryotic and prokaryotic sources (2). These discrepancies can be traced partially to the different assay conditions used and also that in some cases less than fully active enzyme preparations have been used due to the presence of enzyme molecules containing stoichiometric amounts of inhibitory oxaloacetate in tightly bound form at the active site (2, 14, 15). The kinetic properties of both SQR and QFR, however, in general show that fumarate reductases rapidly reduce fumarate in the presence of electron donors of low redox potential and at a somewhat slower rate oxidize succinate in the presence of electron acceptors of high redox potential. By contrast, succinate dehydrogenase shows the opposite effect of the redox potential of electron donors, i.e., higher potential quinones demonstrating higher rates of succinate oxidation. Specific inhibitors of the reaction of eukaryotic SQR with quinones, such as thenoyltrifluoroacetone (TTFA) (16) and the carboxanilides (15, 17), have previously been described; however, the *E. coli* enzymes are notable exceptions in that both are reported as largely resistant to these inhibitors.

The present study was designed to compare the kinetic properties of the quinone-dependent reactions of

both *E. coli* SQR and QFR and to define inhibitors of the quinone-dependent activity. Hydroxylated naphthoquinones such as lapachol and plumbagin have been used as menaquinol analogues in the assay of several *E. coli* anaerobic reductases including QFR (18). These substrates showed high K_m values; however, their high degree of solubility allowed determination of the initial rate of the reaction. The use of more hydrophobic quinones or quinones which show a higher affinity to the enzyme sometimes makes impossible the determination of the kinetic parameters of the reactions. To overcome this problem and to compare directly SQR and QFR activity, a simple assay system recently developed for use with bovine SQR utilizing NADH, NAD(P)–quinone reductase (DT-diaphorase), and ubiquinone (19) was used to measure the quinol–fumarate reductase reaction of the enzymes. Additionally, the coupling of the succinate–quinone reductase reaction of SQR and QFR with the quinol oxidase activity of *E. coli* *bo*-oxidase (20) was used in order to follow the entire quinone-mediated succino-oxidase reaction of the enzymes by following the formation of fumarate at 240 nm. In this communication the results obtained using these assay systems and relative affinities of the binding site to quinol/quinone and quinone analogues in the direct and reverse reactions of SQR and QFR are reported.

MATERIALS AND METHODS

Growth conditions for *E. coli* expressing SQR and QFR. *E. coli* strain DW35 (9) containing a deletion of the *frdABCD* operon ($\Delta frdABCD$) and the SQR operon (*sdhCDAB*) inactivated by an insertional mutation in *sdh* (*sdhC::kan*) was used as the host for expression of fumarate reductase and succinate dehydrogenase. *E. coli* strain GV141, a *recA* derivative of DW35, was also used as host strain for aerobic expression of SQR encoded from plasmid pSDH15 (*sdhC⁺D⁺A⁺B⁺Amp^r*) (13), whereas QFR was expressed anaerobically from plasmid pH3 (*frdA⁺B⁺C⁺D⁺Amp^r*) (6). To obtain QFR cells transformed with pH3 were grown overnight in Luria–Bertani medium with appropriate antibiotics and then a 1:500 dilution was used as an inoculum for anaerobic glycerol–fumarate medium enriched with 0.2% (w/v) tryptone and 0.1% (w/v) yeast extract and appropriate antibiotics as previously described (12). To express SQR cells transformed with pSDH15 were grown aerobically in Luria–Bertani medium supplemented with 50 μ M $FeSO_4$ and 100 μ g/ml ampicillin and harvested in late exponential phase as previously described (21).

Preparation and purification of enzyme complexes. The SQR and QFR complexes were extracted from membranes of *E. coli* prepared as previously described (12) using the nonionic detergent Thesit (Boehringer-Mannheim, Indianapolis, IN). The SQR complex was purified according to published procedures (21) using DEAE fast-flow chromatography and the QFR complex also was purified using DEAE fast-flow chromatography as described (12). Histidine-tagged cytochrome bo_3 -oxidase from *E. coli* used in enzyme activity assays was purified on a Qiagen Ni^{2+} –NTA resin affinity column as previously described (20). Plasmid pJR-HisA encoding the cytochrome bo_3 -oxidase was the generous gift of Dr. Robert Gennis of the University of Illinois, Urbana–Champaign. This plasmid was transformed into *E. coli* GV141 for expression of the bo_3 -oxidase by standard procedures (22). DT-diaphorase used for enzyme assays was purified from rat

liver by Cibacron blue affinity chromatography as described previously (23). Enzyme activity was induced in rat liver by intraperitoneal injection with 0.5 ml of 3-methylcholanthrene (8 mg/ml) for 3 consecutive days prior to harvesting the liver.

Assay of SQR and QFR activities. SQR and QFR as isolated are partially deactivated due to the presence of tightly bound oxaloacetate at their active sites and thus it is necessary to activate the enzymes to fully express their succino-oxidase activity (2, 14, 15). To activate the enzymes, SQR and QFR were diluted to 0.5–1.0 mg/ml in 30 mM potassium phosphate (pH 7.4), 0.5% Thesit (w/v), 0.2 mM EDTA, and 10 mM malonate or succinate. The enzyme complexes were activated by incubation for 20 min at 37°C and then kept at 4°C for the duration of the experiment. Enzymatic activity was determined in 1-cm-path-length cuvettes in a Kontron spectrophotometer with 30 mM potassium phosphate (pH 7.8), 0.006% Thesit (w/v), and 0.2 mM EDTA at an assay temperature of 30°C.

The succinate oxidase reaction was measured in the presence of 10 mM succinate in the assay cuvette. The succinate:phenazine ethosulfate (PES) reaction in the presence of dichlorophenolindophenol (DCIP) ($\epsilon^{600} = 21.8 \text{ mM}^{-1} \text{ cm}^{-1}$, pH 7.8) is measured using 1.5 mM PES and 50 μM DCIP as previously described (12, 15). Succinate-ferricyanide reductase activity is determined with 0.45 mM potassium ferricyanide ($\epsilon^{420} = 1 \text{ mM}^{-1} \text{ cm}^{-1}$). Succinate-quinone reductase activity of SQR determined with Wurster's blue was assayed at 612 nm with variable amounts of quinones as previously described (19). Succinate-Q₁ reductase activity of QFR was followed at 278 nm ($\epsilon^{278} = 11.8 \text{ mM}^{-1} \text{ cm}^{-1}$).

The quinol-fumarate reductase reaction was assayed in a coupled system with rat liver DT-diaphorase, with ubiquinone analogues Q₁ and Q₂ and naphthoquinone analogues DMN and menaquinone-1 (MQ₁) as described elsewhere (19). Quinone analogues Q₁ and MQ₁ were generously supplied by Eisai Co. Ltd. (Tokyo, Japan). The assay mixture contained 120 μM NADH, 10 mM fumarate, and 20–80 $\mu\text{g/ml}$ DT-diaphorase and the reaction was followed by the decrease in absorption at 340 nm after the addition of fumarate. The assays were carried out in aerobic cuvettes with ubiquinone derivatives and in aerobic and anaerobic cuvettes with the lower potential menaquinone derivatives. To achieve anaerobiosis 5 mM glucose, 40 $\mu\text{g/ml}$ glucose oxidase, and 3 $\mu\text{g/ml}$ catalase were added to rubber-stoppered cuvettes and incubated for 10 min prior to addition of quinones.

Analytical methods. Protein concentration was determined by the method of Lowry in the presence of 1% (w/v) SDS with bovine serum albumin as standard. To determine SQR concentration the content of heme b_{556} in SQR was determined by the dithionite-reduced minus air-oxidized difference spectrum using $\epsilon^{556} = 22.8 \text{ mM}^{-1} \text{ cm}^{-1}$. The concentration of purified QFR was measured by assaying the content of covalently bound FAD. This was accomplished by destroying the iron-sulfur centers of the enzyme by incubation of the enzyme (10 mg/ml) in 50 mM acetate buffer (pH 5.0) at 37°C for 30 min. FAD content was then determined in 50 mM potassium phosphate (pH 7.0) using $\epsilon^{445} = 12 \text{ mM}^{-1} \text{ cm}^{-1}$ for histidyl-FAD.

RESULTS

A commonly used method for the succinate-quinone reductase reaction of SQR/QFR is to couple the reduction of quinone by succinate with the chemical oxidation of formed quinol by DCIP as the final electron acceptor (2, 3). DCIP, however, demonstrates a significant low second-order rate constant for quinol oxidation. Thus, electron transfer from quinol to DCIP, especially at low concentrations of quinone, is rate limiting, complicating quantitative kinetic analysis (25).

E. coli SQR and QFR also show complex kinetics of reduction of DCIP (data not shown) as has been described for bovine SQR (24). Wurster's blue, a semiquinonediimine radical of *N,N,N',N'*-tetramethyl-*p*-phenylenediamine, has been used for measurement of the succinate-quinone reductase activity of bovine SQR (25). Unfortunately, although Wurster's blue could be used to assay *E. coli* SQR activity, it could not be used for QFR because purified QFR is highly reactive toward this artificial one-electron acceptor (data not shown). Thus, quinone-mediated assay of QFR/SQR was done by the coupled assays described under Materials and Methods.

Catalytic activity of QFR and SQR. Figure 1A shows a representative example of the succino-oxidase reaction catalyzed by QFR in the presence of excess *bo*-oxidase. As can be seen from the trace, there is very little background activity when only succinate is added to the cuvette while addition of quinone initiates oxidation of succinate by the enzyme. Also shown in the figure is that the reaction of QFR is, as expected, completely sensitive to the active-site inhibitor malonate. Although not shown in Fig. 1A, SQR demonstrates the same activity profile as QFR and the same sensitivity to malonate. In contrast to the sensitivity of both enzymes to malonate, QFR but not SQR is also sensitive to the inhibitor HQNO. The low rate of succinate oxidation observed in the absence of added quinone suggests that the purified enzymes contain very low levels of endogenous quinones. Therefore, it is possible to use fixed concentrations of added quinones to determine the standard kinetic parameters of the reaction. Both QFR (Fig. 1B) and SQR (Fig. 1C) in the succino-oxidase reaction show hyperbolic dependence on quinone concentration. The apparent K_m and V_{max} values obtained from the *bo*-oxidase assay for SQR are the same as results obtained on quinone-mediated Wurster's blue reduction (Fig. 1C). Due to the excess amount of *bo*-oxidase used for the assay, quinones during the reaction remain fully oxidized and even at the maximum concentration their absorbance did not exceed 0.4 at 240 nm. Thus, quinone absorbance did not interfere with measurement of fumarate production at 240 nm. It should be noted that because both the *bo*-oxidase- and DT-diaphorase-coupled systems operate with quinones, they also could be sensitive to the same inhibitors as SQR and QFR. Continuous regeneration of quinones, however, can still be accomplished by increasing the amount of DT-diaphorase or *bo*-oxidase added to the assay system which overcomes any inhibition that may occur.

Both SQR and QFR show simple hyperbolic kinetics for succinate oxidation when operating with either naphthoquinones or benzoquinones (Table I). The enzymes show little specificity between the two classes of

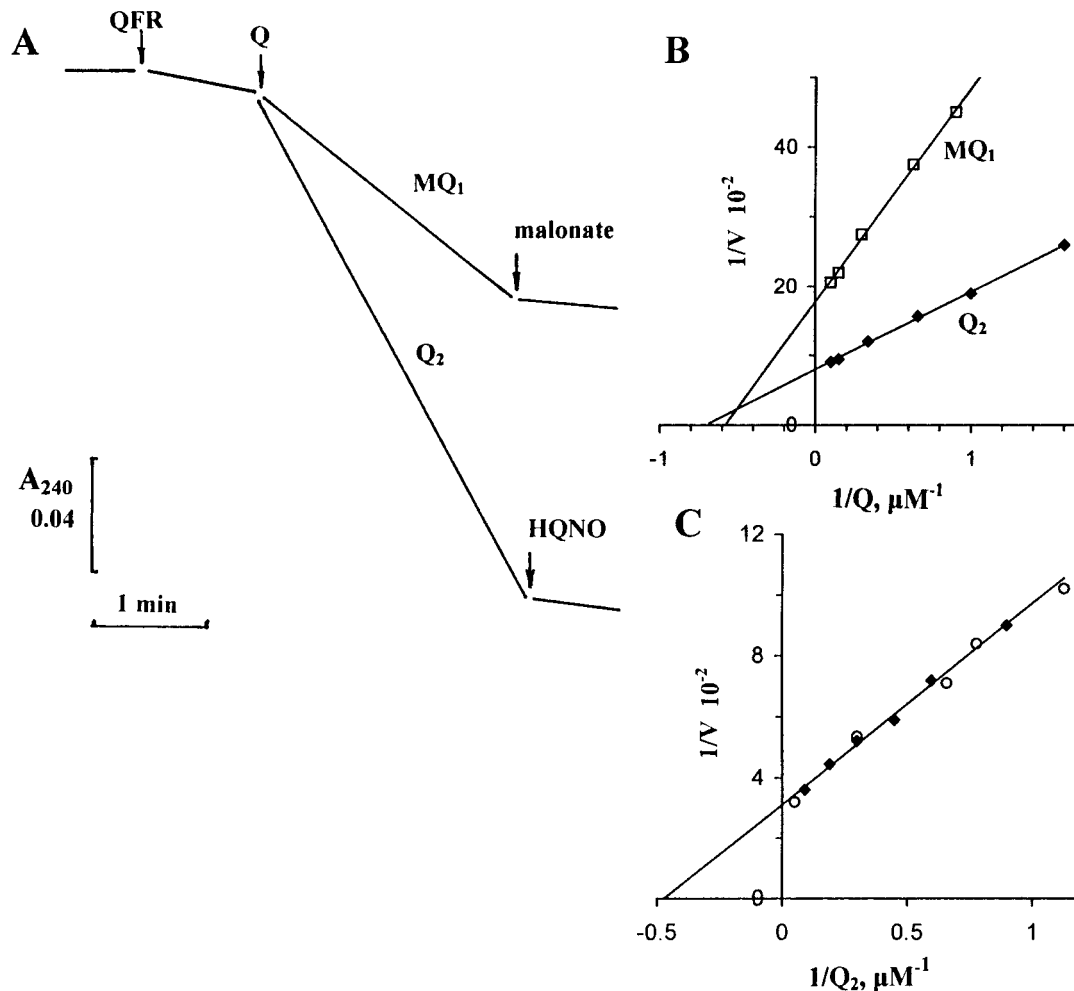


FIG. 1. Catalytic activity of QFR and SQR with quinones. (A). The succino-oxidase reaction catalyzed by coupling operation of QFR and *bo*-oxidase measured by following fumarate production at 240 nm. QFR (2.5 μ g/ml) was placed in a 1-ml cuvette containing 30 mM potassium phosphate, pH 7.8 (30°C), 0.2 mM EDTA, 0.006% (w/v) Thesit, 5 mM succinate, and 20 μ g/ml *bo*-oxidase. MQ₁ or Q₂ at 10 μ M, 10 mM malonate, and 2 μ M HQNO were added as indicated. The absorbance of added quinones at 240 nm did not exceed 0.25. (B). Dependence of succinate oxidase activity of QFR in the presence of *bo*-oxidase on concentration of added Q₂ (◆) and MQ₁ (□). (C). Double-reciprocal plot of succinate-Q₂ reductase activity of SQR determined with Wurster's blue (○) or *bo*-oxidase (◆).

quinones operating with either quinone. Interestingly, the more hydrophobic quinones (Q₂ and MQ₁) show about a one order lower K_m with QFR in succino-oxidase reactions than they do with the more hydrophilic (Q₁ and DMN) derivatives. In contrast, SQR does not show a difference in the K_m value with any of the quinones tested. The turnover rates of forward and reverse reactions of QFR and SQR are dependent upon the redox potential of the quinone. The ability of SQR to reduce fumarate is approximately 2 to 3 times higher with low-potential menaquinols such as DMN and MQ₁ ($E_m = -74$ mV) than with higher potential ubiquinones such as Q₁ and Q₂ ($E_m = +100$ mV). This agrees with data for bovine (26) and *Bacillus subtilis* (27) SQR. The succinate oxidase reaction rate of QFR is more than 2-fold higher with ubiquinones (Table I).

The data in Table I also show that SQR catalyzes succinate oxidation some 40 times faster with ubiquinone than it does fumarate reduction by ubiquinol (either Q₁H₂ or Q₂H₂). The opposite effect is seen with QFR, i.e., it catalyzes fumarate reduction some 12-fold faster than it oxidizes succinate with the lower potential menaquinone derivatives.

Inhibitors of quinone reactions of QFR and SQR. It has been recognized for a number of years that semiquinone (Q[•]) species generated during the reaction cycle of respiratory proteins are stabilized by protein components in the membrane (28). Structures of quinone binding sites have been determined by X-ray crystallography (29, 30). Affinity labels such as azido-quinones and inhibitor-resistant and site-directed mu-

TABLE I

Kinetic Parameters of the Succinate–Quinone and Quinol–Fumarate Reductase Reactions of SQR and QFR

	SQR				QFR			
	Succinate oxidation		Fumarate reduction		Succinate oxidation		Fumarate reduction	
	K_m	k_{cat}^a	K_m	k_{cat}^b	K_m	k_{cat}	K_m	k_{cat}
Q_1/Q_{1H_2} (μM)	2	87	5	1.7	70 ^d	27		
Q_2/Q_2H_2 (μM)	2 ^c	85 ^c	4.5	1.7	1.3	28		
DMN/DMNH ₂ (μM)			4.5	2.4	40	13	50	160
MQ ₁ /MQ ₁ H ₂ (μM)			3	3.4	1.5	14	5.4	177

Note. Assay conditions: 30 mM potassium phosphate, 0.006% Thesit (pH 7.8), 30°C.

^a k_{cat} (s^{-1}) were calculated by heme b_{556} concentration.

^b k_{cat} (s^{-1}) were calculated by FAD concentration.

^c Determined in the *bo*-oxidase and the Wurster's blue-dependent reactions.

^d Determined in the succinate– Q_1 reductase reaction.

tants of respiratory proteins have also been used to define sites where quinones interact (9, 31–34).

There is a large family of inhibitors that are natural and synthetic analogues of quinones that have been used as convenient tools for studies on the mechanism of electron transport. HQNO is an effective and potent ubiquinone and menaquinone antagonist acting on many respiratory cytochrome *b*-containing quinone-reactive redox enzymes (35), as well as those lacking heme such as dimethyl sulfoxide reductase (36) and QFR (37, 38). As seen in Fig. 1A, HQNO is a powerful inhibitor of *E. coli* QFR, although this enzyme is lacking the *b* heme component present in most Complex

IIs. In Fig. 2 it may be seen that both the succinate–quinone and quinol–fumarate reductase reactions of QFR are inhibited by HQNO according to a noncompetitive mechanism. The inhibitor completely blocked MQ₁H₂ oxidation at 2 μM , as well as reduction of Q_2 and MQ₁ by succinate. As shown in Table II the K_i value determined for HQNO of QFR in the fumarate reductase reaction is 0.2 μM which is some three times higher than the K_i for the reverse reaction (succinate oxidation). Succinate oxidations by QFR measured with Wurster's blue, ferricyanide, or phenazine ethosulfate are not inhibited by HQNO. The PES–DCIP activity of the soluble FrdAB domain is also not inhib-

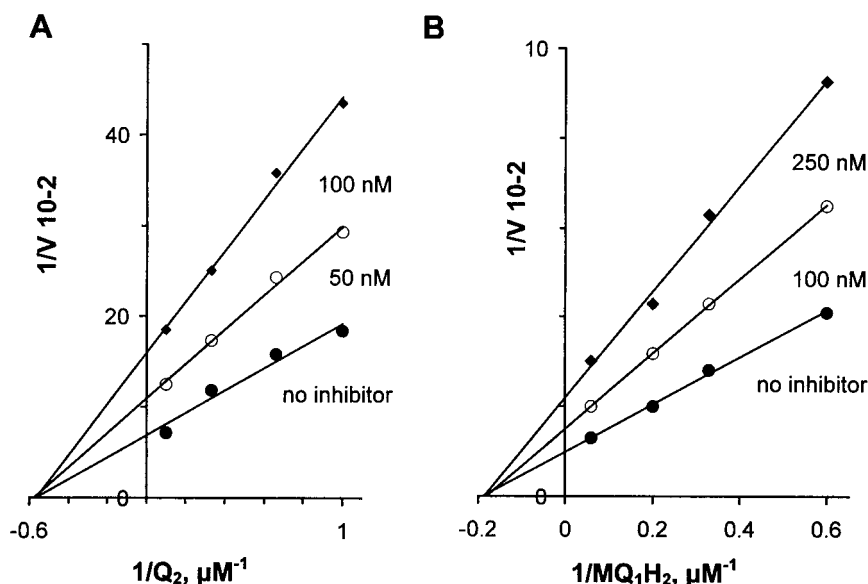


FIG. 2. Inhibition of succinate oxidation and fumarate reduction by QFR with HQNO. (A). Double-reciprocal plot of succino-oxidase activity as in Fig. 1. (B) Fumarate reduction measured in the presence of the indicated concentrations of inhibitor HQNO.

TABLE II

Kinetic Parameters of Inhibition of SQR and QFR (pH 7.8; 30°C)

	SQR		QFR	
	Succinate oxidation	Fumarate reduction	Succinate oxidation	Fumarate reduction
K_i^{HQNO} (μM) ^a	75 μM , not inhibited		0.075	0.2
$K_i^{\text{DNP-17}}$ (μM) ^b	17	25	1.3	2
$K_i^{\text{DNP-20}}$ (μM) ^b	19	27	1.5	2
K_i^{PCP} (μM) ^b	13	17	23	37
$K_i^{\text{MQ}_1}$ (μM) ^b	12			
$K_i^{\text{Q}_2\text{H}_2}$ (μM) ^b				15
K_i^{carboxin} (μM) ^c	30	35	40 μM , not inhibited ^c	

Note. Conditions as in Table I.

^a Calculated form Dixon plots for simple noncompetitive inhibition.

^b Calculated form Dixon plots for competitive inhibition.

^c Mixed-type inhibition, calculated form Dixon plots.

ited by HQNO (data not shown). Although the inhibitor sensitivities of QFR and SQR are similar in most respects, the succinate–Q₂ reductase reaction of *E. coli* SQR, assayed in the presence of DCIP, is not sensitive to inhibition by HQNO even at 75 μM concentration (Table II).

A series of 2-alkyl-4,6-dinitrophenols derivatives have been shown to interfere with quinone reactions of mammalian Complexes I, II, and III but with different potencies and kinetic characteristics (39). Recently it was shown that two of these dinitrophenol derivatives termed 17 and 20 inhibited bovine SQR and *E. coli* QFR according to a noncompetitive mechanism with complex biphasic kinetics in the succinate oxidase reaction (40). A significant difference in K_i values for

succino-oxidase and fumarate reductase reactions was observed. The compounds appeared as better inhibitors in the physiological preferable direction for both the bovine and *E. coli* enzymes. Using the DT-diaphorase and *bo*-oxidase assay systems, we investigated the inhibition pattern of dinitrophenol derivatives 17 and 20 with both *E. coli* QFR and SQR. As seen in Fig. 3, compounds 17 and 20 inhibited the fumarate reduction activity of *E. coli* SQR and QFR according to a competitive mechanism. A similar inhibition pattern is seen for the forward assays of both enzymes (data not shown). The inhibitors rapidly equilibrated with the enzymes, did not show time-dependent changes in activity, and completely blocked the reaction of the enzymes with quinones at high concentrations. As seen in

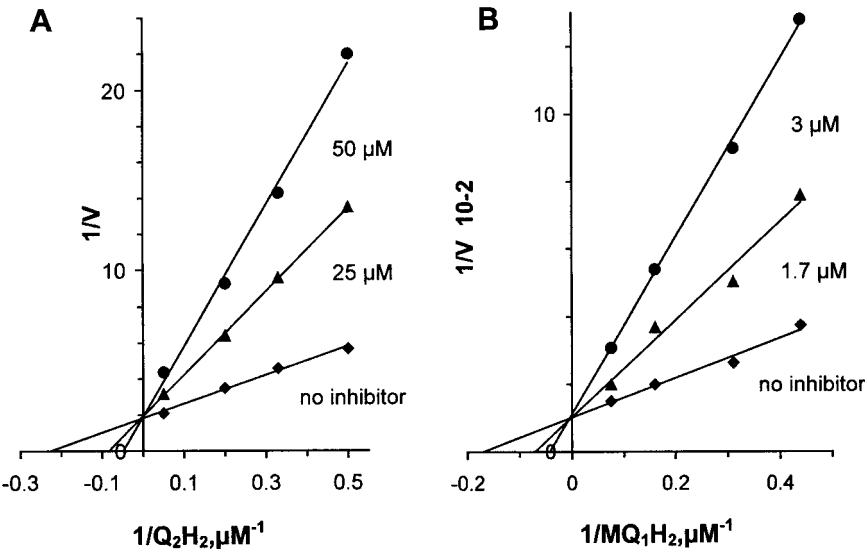


FIG. 3. Double-reciprocal plot of inhibition of fumarate reduction of SQR (A) and QFR (B) by DNP-17. Inhibitor concentrations are indicated. DNP-17 (50 μM) gave an absorbance of 0.6 at 340 nm.

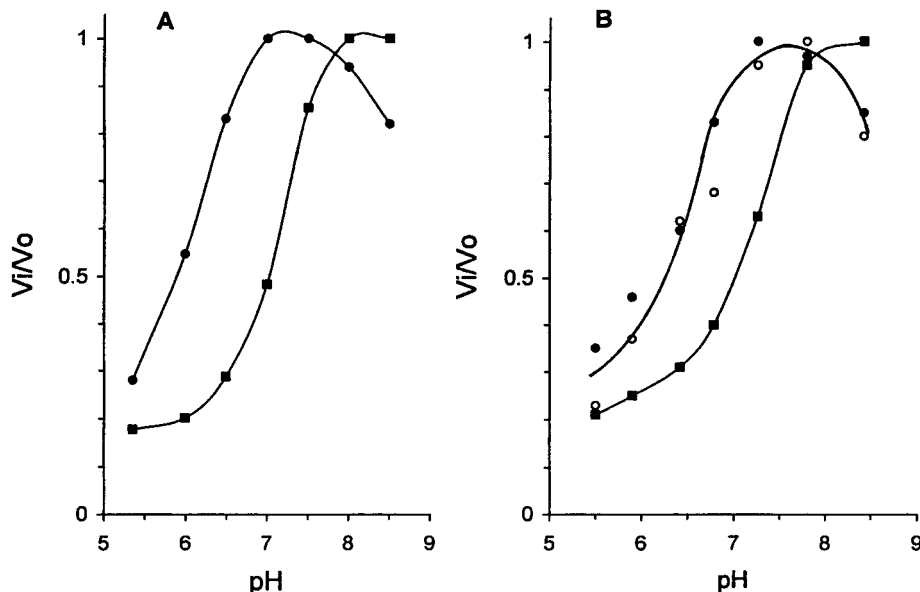


FIG. 4. pH dependence of succinate-quinone reductase and quinol-fumarate reductase activities of QFR (A) and SQR (B). Succinate-quinone reductase activity (■) was determined as described in the legend to Fig. 1 in the presence of 12 μ M Q_2 , 10 mM succinate, and *bo*-oxidase. One hundred percent activity corresponds to 33 and 13 μ mol/min mg protein for SQR and QFR, respectively. Quinol-fumarate reductase activity of SQR (B) was measured in the presence of 12 μ M MQ_1H_2 (●) or Q_2H_2 (○), 10 mM fumarate, and DT-diaphorase. One hundred percent activity corresponds to 1.7 and 0.6 μ mol/min mg protein, respectively. The quinol-fumarate reductase reaction of QFR (A) was determined with MQ_1H_2 as in A with 100% activity at 90 μ mol/min mg protein. The reactions with MQ_1 are carried out under anaerobic conditions in the presence of glucose, glucose oxidase, and catalase, as described under Materials and Methods.

Table II, the K_i for either DNP derivative was one order of magnitude lower for QFR than *E. coli* SQR (2 to 25 μ M). In contrast to the previous results using a different assay system (40) there was not a significant difference in the K_i values for *E. coli* QFR or bovine SQR when assayed in the forward or reverse direction. The data in Table II also show that MQ_1 can act as a competitive inhibitor of the succinate oxidation reaction of SQR with a K_i of 12 μ M and that the ubiquinol analogue Q_2H_2 acts as a competitive inhibitor of fumarate reduction by QFR. These results demonstrate that the catalytic sites for ubi- and menaquinones in *E. coli* SQR and QFR can function with either type of quinone in agreement with the results of physiological studies (11, 12).

Also shown in Table II are the effects on *E. coli* SQR and QFR of the classical inhibitor of bovine SQR carboxin (5,6-dihydro-2-methyl-1,4-oxathiin-3-carboxanilide) (41, 42) and also pentachlorophenol (PCP). Table II indicates that carboxin inhibits *E. coli* SQR with a K_i near 30–35 μ M; however, the inhibition is of a mixed type and this K_i is some 30-fold higher than that found for eukaryotic Complex IIs. In contrast, *E. coli* QFR is not sensitive to carboxin even at 40 μ M concentration. Both *E. coli* enzymes appeared to be inhibited by PCP according to a competitive mechanism with similar K_i values. Known inhibitors of quinone reactions of the

bc₁ complex, such as antimycin, myxothiazol, and stigmatellin, were also tested against SQR and QFR; however, they did not show significant inhibition of the *E. coli* enzymes at concentrations below 30–50 μ M (data not shown).

The pH dependence of fumarate reductase and succinate-oxidase reactions of SQR and QFR. The succinate oxidase and fumarate reductase activities of bovine SQR have been shown to be strongly pH dependent (43). The SQR and QFR complexes from *E. coli* have a high degree of primary sequence homology with the bovine SQR complex; however, data have not been presented for the pH dependence of the fumarate reductase and succinate dehydrogenase activities of these complexes. In Fig. 4 the pH dependence of the quinol-fumarate reductase and succinate-quinone reductase activities for both *E. coli* SQR and QFR is shown. The succinate-quinone reductase activity increased for both enzymes over the pH range 5.5–8.5. The same pH profiles are obtained when activity was measured by the PES-DCIP assay (data not shown). The pH profile for both assays is consistent with the ionization of a group with pK_{as} of 7.4 and 7.6 for QFR and SQR, respectively. The V_{max} values of the quinol-fumarate reductase reaction of the enzymes using the DT-diaphorase-coupled assay show a bell-shaped de-

pendence with apparent pK_{a1} determined from a logarithmic plot equal to 6.6 for QFR and 6.9 for SQR. However, the $K_m^{MQ_2H_2}$ for QFR appeared to be pH independent over the indicated pH range. The activity of SQR with either Q_2H_2 or MQ_1 as acceptor gave similar pH profiles when normalized to reflect the differences in activity of these two quinones.

DISCUSSION

The kinetic data presented here analyze in parallel the quinone-dependent reactions of both *E. coli* SQR and QFR. Using the same experimental conditions for comparisons of the kinetics of these closely related respiratory complexes clarifies similarities and differences between SQR and QFR. An important observation resulting from these studies is that both enzyme complexes show simple hyperbolic dependence of the fumarate reductase and succinate–quinone reductase activities on the concentrations of added quinones. These data taken together with agreement of $K_i^{Q_1}$ and $K_i^{Q_2H_2}$ values for the forward SQR and reverse QFR reactions suggest that there is a single exchangeable quinone site involved in operation with pool ubiquinones and naphthoquinones. Observations from several laboratories suggest that there are two sites where quinones bind to SQR and QFR (9, 44–46); however, only one of these sites has been suggested to exchange with pool quinones in agreement with the results presented here. Comparisons of K_m values for Q_1 and Q_2 (Table I) indicate that the first two isoprenoid units in the quinone molecule are important for the binding of quinones to QFR but not with SQR, suggesting that this site may be more hydrophobic in QFR.

The data also show both similarities and differences in the properties of inhibitors of quinone-dependent reactions of both enzymes. The kinetic parameters of inhibition of activity show that PCP and the 2-alkyldinitrophenol analogues 17 and 20 competitively inhibit both SQR and QFR. The more hydrophobic dinitrophenol analogues appear to be more potent inhibitors for QFR than for SQR, whereas PCP shows rather similar K_i values for both enzymes. In contrast to previous observations (40), the inhibitors tested here did not show a dramatic difference in the K_i value for the forward and reverse reactions; moreover, the data are consistent with slightly higher K_i parameters for the fumarate reductase reaction for both enzymes. The different results reported here for the DNP derivatives probably are due to the known side reactions of the quinone–DCIP assays used previously (40). The inhibition properties of HQNO also differed for QFR and SQR with the former very sensitive to HQNO and the latter resistant to this inhibitor. It is not known why *E. coli* SQR is resistant to HQNO; however, it may be that subtle differences in the architecture of the quinone

binding site exist for SQR vs QFR. In *B. subtilis* SQR which contains two *b* hemes, HQNO as a structural analogue of MQ has been suggested to interact with a menaquinone binding site near the low-potential heme found in that organism's Complex II (35). This has been interpreted to suggest that HQNO binds near the Q-distal site in that organism. Recent results for *E. coli* QFR, however, show that HQNO perturbs the [3Fe–4S] cluster EPR spectrum (38), suggesting that HQNO inhibits near the Q-proximal (quinone binding site near the [3Fe–4S] cluster) site in QFR. Thus, HQNO may bind to topographically different but perhaps functionally similar sites in *E. coli* QFR and *B. subtilis* SQR. Rothery and Weiner (38) demonstrated that HQNO binds to QFR in approximately an equimolar ratio compared to the [3Fe–4S] cluster and elicits a significant change in its EPR lineshape and thus proposed a single dissociable binding site for MQH_2 . In the kinetic studies reported here HQNO appeared to show a noncompetitive inhibition pattern for QFR in both the forward and reverse reactions. As an analogue of menaquinone, HQNO could be structurally related to the radical form of this quinone and when bound to the enzyme elicit a perturbation resembling the transient state of the enzyme during catalytic turnover. Recent crystallographic studies on the bc_1 complex (47) show that different subtypes of Q_0 inhibitors, known to cause varied effects on the EPR spectra of the Rieske iron–sulfur cluster and the optical spectra of heme b_L , occupied different subsites within the Q_0 pocket. The recently available X-ray crystal structure of *E. coli* QFR (48) shows two binding sites for quinones, one of which termed Q-proximal is believed to be the exchangeable quinone binding site. As for the bc_1 complex, the kinetically distinguishable mechanisms of inhibition by analogues of quinones for QFR and SQR may reflect the different binding subsites within the Q-proximal binding pocket.

In this report we have also directly compared the pH dependence for QFR and SQR activity. The pH profile of bovine SQR suggested that ionization of a group with a pK_a near 7.0 was important for enzyme activity. These reports showed a mirror-image pH profile for bovine SQR's succino-oxidase and fumarate reductase activities suggesting the participation of an unprotonated group in succinate oxidation and its protonated form in fumarate reduction (43). It was suggested that the imidazole moiety of a histidyl residue was the likely active-site residue that is responsible for the pH profile of the bovine enzyme. In support of these observations, an active-site histidyl residue near the dicarboxylate binding site was subsequently shown to be important for catalytic activity in *E. coli* QFR by site-directed mutagenesis (4). Protonation–deprotonation of a sulfhydryl group has also been suggested to be important for the pH profile behavior of bovine SQR

(19). In contrast to the previous studies reporting a mirror-image pH profile, the latter report found that the quinol–fumarate reductase reaction catalyzed by isolated SQR using the DT-diaphorase assay had the same pH profile as the succinate dehydrogenase activity (19).

E. coli is an example of a neutrophilic bacterium which maintains its intracellular pH in the range of 7.4–7.8, within the range where isolated SQR and QFR show maximum activity in both forward and reverse directions. The pH dependence of succinate–quinone and quinol–fumarate reductase reactions catalyzed by SQR and QFR reported here for the *E. coli* enzymes is more consistent with those reported by Grivennikova *et al.* (19). The pH profile for fumarate reduction of the four-subunit enzyme complex measured with low-potential electron donors (benzylviologen, $E_m = -359$ mV; and phenosafranine, $E_m = -252$ mV) in the reduced dye assay and that found for the soluble domain of the enzyme (*E. coli* FrdAB, bovine SdhAB) correlate with the pH profile in cyclic voltammetry experiments on bovine SQR (49). Thus, these data may reflect electron transfer reactions occurring in the soluble dehydrogenase fragment of the enzyme, whereas the coupled assays used in this study measure properties of the overall quinone dependent reaction. Thus, the similarity of pH profiles of quinol–fumarate reductase reactions of both enzymes may suggest that similar catalytic groups are involved in quinol deprotonation and oxidation.

The results reported here demonstrate that although *E. coli* SQR and QFR are highly specialized for catalysis in different directions for the reaction between succinate–fumarate and quinone–quinol, they show rather similar behavior in quinone-dependent reactions despite their different turnover numbers. Further mechanistic studies and newly available structural information (48) for Complex II will help to resolve the similarities and differences in the two enzyme complexes.

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